

OBJECT ORIENTED PROGRAMMING USING VB.NET

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Recommended Textbooks

1. **Mike Snell & Lars Powers**; "Microsoft Visual Studio 2010 Unleash" Pearson Education, Inc.
2. **Evangelos Petroutsos**, "Mastering Microsoft Visual Basic 2010", Wiley Publishing Inc.

DATA TYPES

- ❑ Datatype is the kind or form or type of data that can be stored in a computer memory. Datatype is a data storage format that can contain a specific type or range of values. When computer programs store data in variables, each variable must be assigned a specific data type. Some examples of data types include integers, floating point, characters, strings, double, bytes, short, long, boolean and arrays etc.
- ❑ Datatype is the collection of data with values having fixed meaning as well as characteristics. Datatype is a classification that specifies which type of value a variable has and what type of mathematical, relational or logical operations that can be perform on it.
- ❑ Unlike humans, a computer does not know the difference between a number (1234) and text (abcd).

DATA TYPES

- ❑ **Data type** is an attribute of data which tells the [compiler](#) or [interpreter](#) how the programmer intends to use the data. Most programming languages support basic data types of integer numbers, Floating-point numbers, characters and Booleans. A data type constrains the values that an expression, such as a variable or a function, can take. This data type defines the operations that can be done on the data, the meaning of the data, and the way values of that type can be stored. A data type provides a set of values from which an object, variable or function can take its values.
- ❑ Datatype is an attribute that tells what kind of data that value can have. Data types define particular characteristics of data used in software programs and inform the compilers about predefined attributes required by specific variables or associated data objects.

DATA TYPES

- ❑ **Data type** is a classification that dictates what a variable or object can hold in computer programming. Data types are an important factor in virtually all computer programming languages, including C#, C++, JavaScript, and Visual Basic. When programmers create computer applications, both desktop and web-based, data types must be referenced and used correctly to ensure the proper result and an error-free program.
- ❑ Some programming languages require the programmer to define the data type of a variable before assigning it a value. Other languages can automatically assign a variable's data type when the initial data is entered into the variable. Note that, every programming language has its own datatype implementations that it can read, understand, reference and use in its programs.

DATA TYPES

- ❑ **Classification of Datatypes:** Datatypes can be classified as follows;
- ❑ **Primitive types:** [Primitive data types](#) are typically types that are built-in or basic to a language implementation. E.g. Machine datatypes, Boolean types, Numeric types
- ❑ **Composite types:** [Composite types](#) are derived from more than one primitive type. Transforming a primitive type into a compound type generally results in a new type, for instance, an *array of integer* is a different type from *integer*. E.g. Arrays, Unions, Records, Sets, Objects, Strings, Enumeration, Pointers.
- ❑ **Abstract datatypes:** Any type that does not specify an implementation is an [abstract data type](#). For instance, a stack can be implemented as an array (a contiguous block of memory containing multiple values), or as a linked list (a set of non-contiguous memory blocks linked by pointers). E.g. Stack, Queue, Set, Tree, Graph, List, Hash etc

DATA TYPES

- ❑ **Classification of Datatypes:**
- ❑ **Character:** A character is a unit of information that roughly corresponds to a symbol such as in an alphabet or number. A character is used to represent a single value.
- ❑ **String:** A string is a sequence of characters, either as a literal constant or as some kind of variable. A string is used to represent multiple characters. Strings are also used to store alphabets and numbers (alphanumeric values)
- ❑ **Integer:** An integer is a positive or negative whole number.
- ❑ **Float:** A float is datatype which is used to store short decimal values or fractional numbers.
- ❑ **Double:** A double is a datatype which is used to store long decimal values or fractional numbers.
- ❑ **Boolean:** Boolean is a [data type](#) that has one of two possible values; a True (1) or False (0) which is used to represent the truth values of [logic](#) gates and [Boolean algebra](#).

DATA TYPES

❑ Classification of Datatypes:

Data Type	Size	Description	Examples
Character	2 bytes	Stores a single character/letter or ASCII values	A, B, C, 1,2,3
String	4 bytes	Stores multiple characters	Alice, Bob, Var12
Integer	4 bytes	Stores whole numbers from -2,147,483,648 to 2,147,483,647	5, 22, 101, 2060
Float	4 bytes	Stores fractional numbers. Sufficient for storing 6 to 7 decimal digits	2.1, 5.25, 7.125
Double	8 bytes	Stores fractional numbers. Sufficient for storing 15 decimal digits	5.102523476842
Boolean	1 bit	Stores true or false values	True (1), False(0)
Byte	1 byte	Stores whole numbers from -128 to 127	25, 44
Short	2 bytes	Stores whole numbers from -32,768 to 32,767	50010, 66000
Long	8 bytes	Stores whole numbers from -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807	80000, 10010000

THE END

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